SET-2

Series AABB1/2

प्रश्न-पत्र कोड
Q.P. Code **56/2/2**

रोल नं.
Roll No.

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें ।
Candidates must write the Q.P. Code on the title page of the answer-book.

- कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ **11** हैं ।
- प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें ।
- कृपया जाँच कर लें कि इस प्रश्न-पत्र में **12** प्रश्न हैं ।
- **कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें ।**
- इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है । प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा । 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे ।
- Please check that this question paper contains **11** printed pages.
- Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains **12** questions.
- **Please write down the serial number of the question in the answer-book before attempting it.**
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

रसायन विज्ञान (सैद्धान्तिक)

CHEMISTRY (Theory)

निर्धारित समय : 2 घण्टे | *अधिकतम अंक : 35*

Time allowed : 2 hours | *Maximum Marks : 35*

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) *This question paper contains **12** questions. **All** questions are compulsory.*

(ii) *This question paper is divided into **three** Sections – **A, B** and **C**.*

(iii) ***Section A** – Questions no. **1** to **3** are very short answer type questions, carrying **2** marks each.*

(iv) ***Section B** – Questions no. **4** to **11** are short answer type questions, carrying **3** marks each.*

(v) ***Section C** – Question no. **12** is case based question, carrying **5** marks.*

(vi) *Use of log tables and calculators is **not** allowed.*

SECTION A

1. Predict the products formed when CH_3CH_2CHO reacts with the following reagents : (Any ***two***) *2×1=2*

(i) PhMgBr and then H_3O^+

(ii) $LiAlH_4$

(iii) HCN

2. Define molar conductivity for the solution of an electrolyte. How does it vary with concentration ? *2*

3. In a reaction $2N_2O_5\,(g) \longrightarrow 4NO_2\,(g) + O_2\,(g)$, the concentration of N_2O_5 decreases from 0·5 mol L^{-1} to 0·4 mol L^{-1} in 10 minutes. Calculate the average rate of this reaction and rate of production of NO_2 during this period. *2*

SECTION B

4. Account for the following : *3×1=3*

(i) Transition metals and their compounds show catalytic activities.

(ii) Zn, Cd and Hg are non-transition elements.

(iii) Zr and Hf are of almost identical atomic radii.

5.

$E^{\ominus}_{M^{2+}/M}$	Cr	Mn	Fe	Co	Ni	Cu	Zn
	– 0·91	– 1·18	– 0·44	– 0·28	– 0·25	+ 0·34	– 0·76

From the given $E^{\ominus}$ values of the first row transition elements, answer the following questions : *3×1=3*

(i) Why is $E^{\ominus}_{Mn^{2+}/Mn}$ value highly negative as compared to other elements ?

(ii) What is the reason for the irregularity in the above $E^{\ominus}$ values ?

(iii) Why is $E^{\ominus}_{Cu^{2+}/Cu}$ value exceptionally positive ?

6. (a) Write three differences between Lyophobic sol and Lyophilic sol. *3*

OR

(b) (i) Write the dispersed phase and dispersion medium of 'milk'.

(ii) What is the cause of Brownian movement in colloidal particles ?

(iii) Why does physisorption decrease with increase in temperature ? *3×1=3*

7. (a) Write the equations involved in the following reactions : *3×1=3*

(i) Cannizzaro reaction

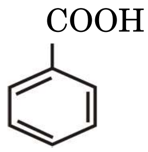
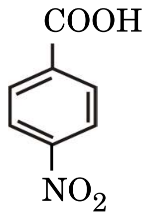
(ii) Aldol condensation

(iii) Hell-Volhard-Zelinsky reaction

OR

(b) (i) Which acid of each pair would you expect to be stronger ? Give reason. *2+1=3*

(I) $F - CH_2 - COOH$ or $Cl - CH_2 - COOH$

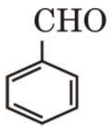
(II)  or 

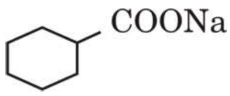
(ii) Distinguish between Propanal and Propanone.

8. A first order reduction takes 30 minutes for 75% decomposition. Calculate $t_{1/2}$. *3*

Given : [log 2 = 0·3, log 3 = 0·48, log 4 = 0·6, log 5 = 0·7]

9. (a) Write the major products in the following : *3×1=3*

(i)  $\xrightarrow[273 - 283\ K]{HNO_3 + H_2SO_4}$

(ii)  $+ NaOH \xrightarrow[\Delta]{CaO}$

(iii) $CH_3 - \underset{\underset{H}{|}}{C} = O \xrightarrow{NH_2OH}$

OR

(b) (i) Oxidation of propanal is easier than propanone. Why ?

(ii) How can you distinguish between Acetophenone and Benzophenone ?

(iii) Draw the structure of the following derivative :

2,4-Dinitrophenylhydrazone of Propanone *3×1=3*

10. Calculate the emf of the following cell : *3*

$$Zn\ (s)\,|\,Zn^{2+}\ (0{\cdot}01\ M)\ ||\ (0{\cdot}001\ M)\ Ag^{+}\,|\,Ag\ (s)$$

Given : $E^{\ominus}_{Zn^{2+}/Zn} = -\,0{\cdot}76\ V$ and

$E^{\ominus}_{Ag^{+}/Ag} = +\,0{\cdot}80\ V$

[log 2 = 0·3010, log 3 = 0·4771, log 10 = 1]

11. (a) Write the hybridisation and magnetic character of the following complexes : *3×1=3*

(i) $[NiCl_4]^{2-}$

(ii) $[Co(NH_3)_6]^{3+}$

(iii) $[FeF_6]^{3-}$

[Atomic number : Ni = 28, Co = 27, Fe = 26]

OR

(b) (i) Write the IUPAC name of the following complex :

$[Co(NH_3)_4(H_2O)Cl]Cl_2$

(ii) What is the difference between an Ambidentate ligand and a Bidentate ligand ?

(iii) Out of $[Fe(NH_3)_6]^{3+}$ and $[Fe(C_2O_4)_3]^{3-}$, which complex is more stable and why ? *3×1=3*

SECTION C

12. Read the following passage and answer the questions that follow : *1+1+1+2=5*

Amines constitute an important class of organic compounds derived by replacing one or more hydrogen atoms of ammonia molecule by alkyl/aryl groups. Amines are usually formed from nitro compounds, halides, amides, etc. They exhibit hydrogen bonding which influences their physical properties. Alkyl amines are found to be stronger bases than ammonia. In aromatic amines, electron releasing and withdrawing groups, respectively increase and decrease their basic character. Reactions of amines are governed by availability of the unshared pair of electrons on nitrogen. Influence of the number of hydrogen atoms at nitrogen atom on the type of reactions and nature of products is responsible for identification and distinction between primary, secondary and tertiary amines. Reactivity of aromatic amines can be controlled by acylation process.

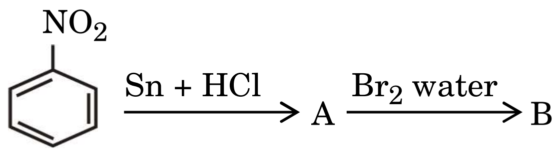
(i) Why does aniline not give Friedel-Crafts reaction ? *1*

(ii) Arrange the following in the increasing order of their pK_b values : *1*

$C_6H_5NH_2$, NH_3, $C_2H_5NH_2$, $(CH_3)_3N$

(iii) How can you distinguish between $CH_3CH_2NH_2$ and $(CH_3CH_2)_2NH$ by Hinsberg test ? *1*

(iv) (a) Write the structures of A and B in the following reactions : *2×1=2*

(I) 

(II) $CH_3CH_2CONH_2 \xrightarrow{Br_2 / \text{alc. KOH}} A \xrightarrow[\text{Pyridine}]{CH_3COCl} B$

OR

(b) How will you convert the following : *2×1=2*

(I) Benzoic acid to aniline

(II) Aniline to p-bromoaniline